

An Optimal Variable Impedance Control With Consideration of the Stability

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Abstract—This letter presents an approach to develop a variable impedance controller with considerations of the optimality and stability. Firstly, an original optimal variable law is designed via demonstration learning, through which the Gaussian mixture model/Gaussian mixture regression (GMM/GMR) algorithm is employed to transfer the human impedance functions to the robot. By using the formulation of the GMR result, the regression task can be completed without using the ground-truth information of the human impedance parameters. To ensure the stability, a minimal complementary input is designed for the learned second-order impedance system. We transform the design problem to a constrained convex optimization problem, of which the constraints are related to a Lyapunov function. A criterion for choosing the Lyapunov functions is presented to ensure the feasibility of the problem, and an analytical solution is computed. The proposed approach is verified by the robotic-assisted rehabilitation and trajectory reproduction experiments conducted on a 7-DOF Franka Panda robot.

Index Terms—Variable impedance control, demonstration learning, stability and optimality.

I. INTRODUCTION

IMPEDANCE control is a basic control architecture for contact-rich robotic tasks [1]. However, constant impedance control may limit impedance control applications in structured environments due to the lack of flexibility. Variable impedance control (VIC) is more consistent with biological machines [2] and can extend the robot skills to more complex tasks. Thus, in recent years, numerous works [3], [4] have been done to explore the VIC benefits.

Existing VIC design methods can be generally classified into two categories. The first one designs time-dependent variable laws. In [5], [6], impedance controllers were incorporated into the dynamic movement primitives framework and optimized by using the policy improvement with path integrals algorithm, which finally generated impedance-parameter trajectories. In [7], damping and stiffness trajectories were obtained by

using a gradient-based method to reduce the trajectory tracking error. The aforementioned methods generate impedance trajectories in an offline manner, which is suitable for structured tasks, such as polish or wipe tasks. However, these methods may fail or lose their optimality when the contacted environments are variable, which is common in unstructured tasks. Some works overcome this issue by online planning the impedance parameters. LQR-based methods are commonly used to generate the optimal impedance parameters online for handling the uncertainty of the contact-force variation [8], [9] or human-demonstrated reference trajectories [10]–[12].

The aforementioned methods, including online and offline methods, only consider the optimality of the variable law. None of them analyzes the stability of the obtained second-order impedance systems. However, stability somehow represents the contact safety which should never be neglected. In [13], [14], tank-based methods were presented to ensure the passivity by suppressing the energy injected by the variable impedance parameters. In [15], a stability condition for varying stiffness parameter was presented by designing a special Lyapunov candidate function.

Though time-dependent variable laws have been shown their effectiveness in some results, they are still not flexible enough to deal with environmental changes since the corresponding impedance systems are non-autonomous. Humans generally modulate their impedance parameters according to the environment state, and this property is inherited by the second type of VIC, of which the variable law is state-dependent. Recent years, reinforcement learning (RL) algorithms are popular to design state-dependent variable laws. According to the complexities of tasks, different RL algorithms can be used, such as classical RL algorithm used in [16] and deep RL (DRL) algorithm used in [17]. To ensure the stability, action space of the DRL algorithm was constrained by a theoretical bound in [18]. One of the main drawbacks of these RL-based methods is that the variable laws are generally trained in simulations, and thus, their practicability are largely influenced by the gaps between simulation and real environments.

Another more intuitive way to design state-dependent variable laws is using regression algorithms. The most representative algorithm may be the Gaussian mixture model/Gaussian mixture regression (GMM/GMR) algorithm due to its stochastic background and concise form [19], [20]. In [21], some constraints were introduced into the GMM optimization step to make the resulted GMR system be stable. However, the constrained optimization problem is very complex and the optimization

Manuscript received September 18, 2021; accepted January 3, 2022. Date of publication January 11, 2022; date of current version January 20, 2022. This letter was recommended for publication by Associate Editor C. Yang and Editor J. Kober upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant 61973275, in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LD21F030002, and in part by the Fundamental Research Funds for the Provincial Universities of Zhejiang RF-A2020004. (Corresponding author: Andong Liu.)

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Digital Object Identifier 10.1109/LRA.2022.3141759

algorithm actually searches a balance between the stability and optimality in practical applications. A control Lyapunov function-based method to ensure the stability of systems learned by general regression algorithms was proposed in [22]. Different with [21], the result in [22] was obtained by solving convex optimization problems which have analytical solutions, and thus is more efficient and practically useful. However, the result was derived for the first-order system and can not be directly used for some second-order systems, such as the impedance system considered in this letter. Another problem for fitting state-dependent variable laws by using regression algorithms is that the ground truth of the human impedance parameters are difficult to be obtained by directly using sensors.

In this letter, we present an approach to design an optimal and stable state-dependent impedance variable law based on GMM/GMR without using ground truth of human impedance parameters. Main contributions are listed as follows:

1) A simple constraint is introduced into the GMM optimization step to make the resulted GMR system be consistent with the impedance system, and hence we can skip the complicated works to obtain the ground-truth information of human impedance parameters. Moreover, the constrained optimization problem is transformed as an unconstrained problem which can be easily solved.

2) To ensure the stability of the learned impedance system, we extend the Lyapunov function-based method presented in [22] for the first-order system to the second-order system, among which we present a criterion for choosing the Lyapunov function to ensure the feasibility of the problem.

3) We propose an approach to design a quadratic Lyapunov candidate function which satisfies the presented criterion. Based on the designed Lyapunov candidate function, we can stabilize the learned impedance system with an optimal input law in analytical form.

The proposed method is validated by various experiments conducted on the 7-DOF Franka Panda robot.

II. PROBLEM STATEMENT

We consider the following m -DOF robot dynamic model

$$H(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = J^T f + \tau \quad (1)$$

where $q \in R^m$ is the joint position, $H(q) \in R^{m \times m}$, $C(q, \dot{q}) \in R^{m \times m}$ and $g(q) \in R^m$ are the inertia matrix, Coriolis/centrifugal matrix, and gravity vector, respectively. Matrix $J(q) \in R^{n \times m}$ is the robot Jacobian, the vector $f \in R^n$ is the contact force and n is the dimension of the work space. In the considered task, the robot is required to learn the human-impedance behaviors by his/her demonstrations. In the demonstration stage, the robot is governed by the following impedance law:

$$M\ddot{x} + D\dot{x} = f \quad (2)$$

where $x \in R^n$ represents the robot position in the work space, $M \in R^{n \times n}$ and $D \in R^{n \times n}$ are the inertia and damping matrices, respectively, and they are both positive definite. The impedance law (2) can be achieved by the following dynamic

control law

$$\tau = HJ^+ \left(M^{-1}(f - D\dot{x}) - \dot{J}\dot{q} \right) + C\dot{q} + g - J^T f \quad (3)$$

where J^+ is the pseudo-inverse of the robot Jacobian J . The human is asked to drag the robot to complete multiple trajectories which are served as the demonstration data set.

In the learning stage, the robot is asked to imitate human impedance behaviors, i.e., to generate suitable forces to reproduce trajectories. Thus, the specific problem is how to establish the relationship between the estimated human force and the current robot state. We establish the following human (limb) impedance model [23]:

$$f = -K_h(e, \dot{e})e - D_h(e, \dot{e})\dot{e} \quad (4)$$

where $e = x - x_d$ represents the motion error, and x_d is the goal position which is known. We can also consider the attitude error by simply expanding the dimensions of the corresponding vectors and matrices in (4). It should be noted that human stiffness K_h and damping D_h are state-dependent, which raises difficulties to directly measure such impedance parameters. Without these measured ground-truth parameters, traditional regression methods, including neural networks, least-squares, etc., are invalid.

Another problem is that we can not ensure the stability of the following system which is obtained by substituting the learned human impedance model (4) in to system (2)

$$M\ddot{e} + (D + D_h(e, \dot{e}))\dot{e} + K_h(e, \dot{e})e = 0 \quad (5)$$

Remark 1: We considered goal-directed motions in this paper, that is $\dot{x}_d = \ddot{x}_d = 0$. Thus, $\ddot{e} = \ddot{x}$ and $\dot{e} = \dot{x}$. Also, the proposed method introduced in Section III can be easily extended to the case where $\dot{x}_d \neq 0$ and $\ddot{x}_d \neq 0$.

Since the formulation of $D_h(e, \dot{e})$ or $K_h(e, \dot{e})$ can be very complex, the stability analyses are difficult to be directly conducted. Motivated by [22], a complementary input $u \in R^n$ is introduced into the system (5) to stabilize it as

$$M\ddot{e} + (D + D_h(e, \dot{e}))\dot{e} + K_h(e, \dot{e})e + Mu = 0 \quad (6)$$

Since the introduction of u will reduce the optimality obtained by the demonstration learning, it is expected to be as small as possible. The minimal complementary input u is computed by solving the following convex optimization problem

$$\min \quad \frac{1}{2}u^T u \quad (7)$$

$$\text{s.t.} \quad \begin{cases} \left(\frac{\partial V}{\partial s} \right)^T (A(s)s + Bu) \leq -\rho(s), s \neq 0 \\ \left(\frac{\partial V}{\partial s} \right)^T (A(s)s + Bu) = 0, s = 0 \end{cases} \quad (8)$$

where

$$\begin{aligned} s &= \begin{bmatrix} e & \dot{e} \end{bmatrix}^T \\ A(s) &= \begin{bmatrix} O_{n \times n} & I_{n \times n} \\ -M^{-1}K_h(e, \dot{e}) & -M^{-1}(D + D_h(e, \dot{e})) \end{bmatrix} \\ B &= \begin{bmatrix} O_{n \times n} & I_{n \times n} \end{bmatrix}^T \end{aligned} \quad (9)$$

Positive definite function $V(s)$ serves as the Lyapunov candidate function and satisfies $V(s) \rightarrow \infty$ when $\|s\| \rightarrow \infty$. If we define $\rho(s)$ as a positive definite function, all feasible solution of the optimization problem (7), (8) can make the system (6) be globally asymptotically stable (GAS) at $s = 0$. However, due to the specific formulation of the matrix B , the feasibility of the above optimization problem is determined by the chosen Lyapunov candidate function $V(s)$. Thus, a criterion for choosing $V(s)$ to raise a feasible optimization problem for any s is the key to design the minimal complementary input.

Concluded by the above discussion, the main purpose of this paper are given as follows

1) To estimate human-impedance functions $K_h(e, \dot{e})$ and $D_h(e, \dot{e})$ without using their ground-truth information.

2) To present a criterion for choosing $V(s)$ which could ensure the feasibility of the optimization problem (7), (8). We also propose an approach to design a quadratic $V(s)$ which satisfies the presented criterion, and derive the minimal u in an analytical form.

III. METHODOLOGY

A. Demonstration Learning for Human Impedance Functions

In this subsection, we present a constrained GMM/GMR approach to learn human-impedance functions $K_h(e, \dot{e})$ and $D_h(e, \dot{e})$ without using their ground-truth information. After human demonstrations, we can obtain a demonstration data set $D = \{s_{t,n}, f_{t,n}\}_{t=1, n=1}^{T_n, N}$.

We assume that (s, f) follows the GMM as

$$\begin{aligned} p(s, f) &= \sum_{k=1}^K p(k)p(s, f|k) \\ p(s, f|k) &= N\left(\begin{bmatrix} s & f \end{bmatrix}^T \middle| \mu^k, \Sigma^k\right) \\ \mu^k &= \begin{bmatrix} \mu_s^k & \mu_f^k \end{bmatrix}^T \\ \Sigma^k &= \begin{bmatrix} \Sigma_{ss}^k & \Sigma_{sf}^k \\ \Sigma_{fs}^k & \Sigma_{ff}^k \end{bmatrix} \end{aligned} \quad (10)$$

where K is the number of the Gaussian components, k denotes the index of the Gaussian component, μ^k and Σ^k represent the mean value and covariance matrix of the k^{th} Gaussian component, respectively. By denoting $p(k) = \pi^k$, all parameters needed to trained are $\theta = \{\pi^1, \dots, \pi^K, \mu^1, \dots, \mu^K, \Sigma^1, \dots, \Sigma^K\}$. Given an input s , the posterior distribution $p(f|s)$ can be computed as

$$\begin{aligned} p(f|s) &= \sum_{k=1}^K p(k|s)p(f|s, k) \\ p(k|s) &= \frac{p(s|k)p(k)}{\sum_{\bar{k}=1}^K p(s|\bar{k})p(\bar{k})} \\ p(f|s, k) &= N(f|\mu_{f|s,k}, \Sigma_{f|s,k}) \\ \mu_{f|s,k} &= \mu_f^k + \Sigma_{fs}^k (\Sigma_{ss}^k)^{-1} (s - \mu_s^k) \end{aligned} \quad (11)$$

By denoting $h^k(s) = p(k|s)$, the GRM result $p(f|s)$ can be computed as

$$\begin{aligned} \mu_{f|s} &= \sum_{k=1}^K h^k(s) (A^k s + b^k) \\ A^k &= \Sigma_{fs}^k (\Sigma_{ss}^k)^{-1} \\ b^k &= \mu_f^k - A^k \mu_s^k \end{aligned} \quad (12)$$

Compared the formulations of the above GMR result and the system (4), it is not difficult to find that these two formulations will be consistent if we constrain $b^k = 0$. The estimated human-impedance model will be

$$\begin{aligned} f &= \mu_{f|s} = \sum_{k=1}^K h^k(s) A^k s \\ &= \begin{bmatrix} -K_h(e, \dot{e}) & -D_h(e, \dot{e}) \end{bmatrix} s \end{aligned} \quad (13)$$

Parameters θ is obtained by solving the following constrained GMM optimization problem

$$\min_{\theta} J(\theta) = -\frac{1}{T_n N} \sum_{n=1}^N \sum_{t=1}^{T_n} \log(p_{\theta}(s_{t,n}, f_{t,n})) \quad (14)$$

subject to

$$\begin{cases} b^k = 0 \\ \Sigma^k \succ 0 \\ 0 < \pi^k < 1 \\ \sum_{i=1}^I \pi^k = 1 \end{cases}$$

Remark 2: Constraints $b^k = 0, k = 1, \dots, K$ considered in [21] were used for stabilization purpose. In this letter, they are used to ensure the consistency between the GMR result (12) and the system (4), and hence we can skip the complicated works to obtain the ground-truth information of human impedance parameters.

The above optimization problem can be approximated by an unconstrained problem after using the variable-change trick. Define the new parameters as $\bar{\theta} = \{\omega_1, \dots, \omega_K, L_1, \dots, L_K, \mu_s^1, \dots, \mu_s^K\}$ such that

$$\begin{cases} \pi^k = \frac{e^{\omega_k}}{\sum_{k=1}^K e^{\omega_k}} \\ \Sigma^k = L_k L_k^T + \delta I \\ \mu_f^k = A^k \mu_s^k \end{cases} \quad (15)$$

where L_k is a lower triangular matrix, and $\delta \in R^+$. We can obtain the following unconstrained optimization problem which is approximately equivalent with (14)

$$\min_{\bar{\theta}} J(\bar{\theta}) = -\frac{1}{T_n N} \sum_{n=1}^N \sum_{t=1}^{T_n} \log(p_{\bar{\theta}}(s_{t,n}, f_{t,n})) \quad (16)$$

where $p_{\bar{\theta}}(s_{t,n}, f_{t,n})$ is obtained by substituting new parameters into $p_{\theta}(s_{t,n}, f_{t,n})$ according to (15). It should be noted that the gap between optimization problems (16) and (14) is caused by the term $\Sigma^k = L_k L_k^T + \delta I$ where we fix a small enough δ to ensure the positive definite property.

Remark 3: In this subsection, we consider the case when $\dot{x}_d = 0$ and thus the component $\dot{e}$ in s equals to the velocity $\dot{x}$ which can be directly collected by using the robot sensor. However, if $\dot{x}_d \neq 0$, we can still obtain the component $\dot{e}$ by computing it as $\dot{e} = \dot{x} - \dot{x}_d$ since $\dot{x}_d$ is always known for us. The value of $\ddot{x}_d$ will not influence the approach proposed in this subsection. Thus, we claim that the approach proposed in this subsection still works for the case $\dot{x}_d \neq 0$ and $\ddot{x}_d \neq 0$.

B. Minimal Complementary Input Design

In this subsection, we firstly present a criterion for designing the Lyapunov candidate function V to ensure the feasibility of the convex optimization problem (7), (8). Then, we propose an approach to design a typical quadratic V which satisfies the presented criterion. Finally, we compute the analytical solution u for the convex optimization problem (7), (8) with the designed quadratic V .

Proposition 1: The optimization problem (7), (8) is feasible for any s if and only if when s satisfies $B^T \frac{\partial V}{\partial s} = 0$, the following inequality holds.

$$\left(\frac{\partial V}{\partial s}\right)^T A(s)s + \rho(s) \leq 0 \quad (17)$$

Proof: When $s \neq 0$, the feasibility of the optimization problem is determined by the feasibility of the inequality equation

$$\left(\frac{\partial V}{\partial s}\right)^T Bu \leq -\rho(s) - \left(\frac{\partial V}{\partial s}\right)^T A(s)s \quad (18)$$

Thus, when s satisfies $\frac{\partial V}{\partial s}^T B \neq 0$, we will always have feasible solutions. Otherwise, if s satisfies $B^T \frac{\partial V}{\partial s} = 0$, the left side is always 0, and the inequality is hold only if $\rho(s) + \left(\frac{\partial V}{\partial s}\right)^T A(s)s \leq 0$.

When $s = 0$, the optimization problem (7), (8) is always feasible, e.g., $u = 0$. And in this case, the inequality (17) is always hold, which completes the proof. ■

Directly designing the Lyapunov candidate function V according to the criterion presented in the Proposition 1 is difficult. We here introduce an approach to design a typical quadratic V .

Proposition 2: A pair of functions $(V(s), \rho(s))$ which satisfies the criterion proposed in Proposition 1 can be designed as

$$\begin{cases} V(s) = s^T P s \\ \rho(s) = \alpha s^T Q s \end{cases} \quad (19)$$

where constant $\alpha \in (0, 1]$, Q is any positive-definite matrix, and P is positive-definite matrix obtained by solving the following matrix equation

$$P\bar{A} + \bar{A}^T P = -Q \quad (20)$$

where $\bar{A}$ is the matrix following the form

$$\bar{A} = \begin{bmatrix} 0_{n \times n} & I_{n \times n} \\ -\bar{K}_{n \times n} & -\bar{D}_{n \times n} \end{bmatrix} \quad (21)$$

and $\bar{K}$, $\bar{D}$ are matrices such that $\bar{A}$ is Hurwitz.

Proof: Since $\bar{A}$ is Hurwitz, we can compute a positive-definite matrix P by solving (20) for any positive-definite matrix Q . Consider the Lyapunov candidate function $V(s) = s^T P s$, its derivative along the system $\dot{s} = \bar{A}s$ can be computed as

$$\dot{V} = s^T (P\bar{A} + \bar{A}^T P)s = -s^T Q s \leq 0 \quad (22)$$

Consider the criterion proposed in Proposition 1, for any s satisfying $B^T \frac{\partial V}{\partial s} = 0$, the formulation of $\frac{\partial V}{\partial s}$ must be

$$\left(\frac{\partial V}{\partial s}\right)^T = \left[\left(\frac{\partial V}{\partial e}\right)^T, 0_{1 \times n}\right] \quad (23)$$

Then, we have

$$\begin{aligned} \left(\frac{\partial V}{\partial s}\right)^T A(s) &= \left(\frac{\partial V}{\partial s}\right)^T \bar{A} \\ &= \left[0_{1 \times n}, \left(\frac{\partial V}{\partial e}\right)^T\right] \end{aligned} \quad (24)$$

due to the special formulations of $A(s)$ and $\bar{A}$. Thus, for any s satisfying $B^T \frac{\partial V}{\partial s} = 0$, we have

$$\begin{aligned} \left(\frac{\partial V}{\partial s}\right)^T A(s)s &= \left(\frac{\partial V}{\partial s}\right)^T \bar{A}s \\ &= \dot{V} \\ &= -s^T Q s \leq 0 \end{aligned} \quad (25)$$

If we set $\rho(s) = \alpha s^T Q s$, $\alpha \in (0, 1]$ we can obtain

$$\left(\frac{\partial V}{\partial s}\right)^T A(s)s + \rho(s) \leq 0 \quad (26)$$

which completes the proof. ■

Based on the Lyapunov function V designed in Proposition 2, we can derive a complementary input u which can make the system (6) be GAS at $s = 0$ by analytically solving the convex optimization problem (7), (8).

Theorem 1: The complementary input

$$u = \begin{cases} 0, 2s^T P A(s)s + \alpha s^T Q s \leq 0 \\ -\frac{2s^T P A(s)s + \alpha s^T Q s}{2s^T P B B^T P s} B^T P s, \text{ else} \end{cases} \quad (27)$$

can make the system (6) be GAS at $s = 0$, if $\alpha \in (0, 1]$, P , Q are constructed according to the Proposition 2.

Proof: The KKT conditions of the convex optimization problem (7), (8) are

$$\begin{cases} \left(\frac{\partial V}{\partial s}\right)^T A(s)s + \rho(s) + \left(\frac{\partial V}{\partial s}\right)^T B u \leq 0 \\ \lambda \geq 0 \\ \lambda \left(\left(\frac{\partial V}{\partial s}\right)^T A(s)s + \rho(s) + \left(\frac{\partial V}{\partial s}\right)^T B u \right) = 0 \\ u + \lambda B^T \frac{\partial V}{\partial s} = 0_3 \end{cases} \quad (28)$$

where λ is the dual variable. By solving the KKT conditions, we can obtain the following solution

$$u = \begin{cases} 0, \left(\frac{\partial V}{\partial s}\right)^T A(s)s + \rho(s) \leq 0 \\ -\frac{\left(\frac{\partial V}{\partial s}\right)^T A(s)s + \rho(s)}{\left(\frac{\partial V}{\partial s}\right)^T B B^T \frac{\partial V}{\partial s}} B^T \frac{\partial V}{\partial s}, \text{ else} \end{cases} \quad (29)$$

If the Lyapunov candidate function V satisfies the Proposition 1, then we can conclude that $\left(\frac{\partial V}{\partial s}\right)^T A(s)s + \rho(s) > 0$ implies

$B^T \frac{\partial V}{\partial s} \neq 0$, which ensure the rationality of the second row of the above solution. By substituting V, ρ designed in Proposition 2 into the above solution, we can obtain

$$u = \begin{cases} 0, 2s^T P A(s)s + \alpha s^T Q s \leq 0 \\ -\frac{2s^T P A(s)s + \alpha s^T Q s}{2s^T P B B^T P s} B^T P s, \text{ else} \end{cases} \quad (30)$$

Thus, the complementary input u is optimal, which implies it can make the system (6) be GAS at $s = 0$ and completes the proof. ■

Remark 4: In this paper, we consider the case when $\ddot{x}_d = \dot{x}_d = 0$ to simplify the formulation of the convex optimization problem (7), (8). If $\dot{x}_d \neq 0$ and $\ddot{x}_d \neq 0$, the input u in constraints (8) should be replaced by $\bar{u} = u + u_d$ where $u_d = -M^{-1}\ddot{x}_d - D\dot{x}_d$. In this case, the results in Proposition 1 and Proposition 2 will not be changed since they do not depend on the specific form of $\bar{u}$. The resulted optimal u in Theorem 1 can still be derived by solving corresponding modified KKT conditions. Thus, we claim that the approach still works in the case when $\dot{x}_d \neq 0$ and $\ddot{x}_d \neq 0$.

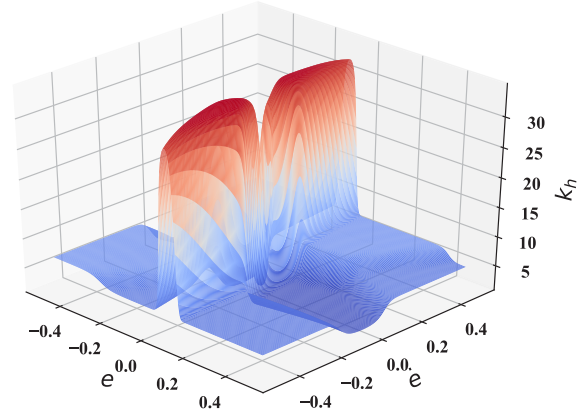
IV. EXPERIMENTS

In this section, two experiments are conducted to show the effectiveness of the proposed method. The first experiment is a robotic-assisted rehabilitation task where the robot is asked to mimic a therapist to provide supported force for a patient in order to complete some motions. The motion is limited in one dimension in order to visualize some results. The second experiment is a trajectory reproduction task where the robot is asked to reproduce some trajectories in three-dimensional space via VIC after human demonstrations.

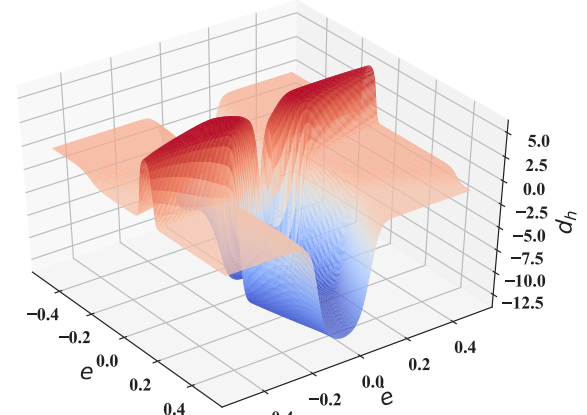
A. Robotic-Assisted Rehabilitation Experiment

In this experiment, a therapist is firstly asked to drag the robot to generate one-dimensional motions, through which we can collect a demonstration data set D . Then, a patient is asked to generate a goal-directed motion, and the robot will simultaneously provide supported force $f_s = f_h + f_c$ where $f_h = -d_h(e, \dot{e})\dot{e} - k_h(e, \dot{e})e$ and $f_c = -mu$ are estimated therapist's force and complementary force, respectively. The functions $d_h(e, \dot{e})$, $k_h(e, \dot{e})$ and u are designed by using the method introduced in Section III. Since the motion is limited in one-dimensional space, the symbols are written as their lowercases. Robot impedance parameters in (2) are set as $m = 1, d = 1$.

The demonstration data set contains 10 goal-directed motions trajectories of $(e, \dot{e}, f_h)$. For GMM/GMR-based demonstration learning, we set the number of Gaussian components $K = 5$. The human-impedance learning results are shown in Fig. 1. It can be found in Fig. 1(a) that the large k_h only appears in the small-error area, which implies that the spring-like force dominates the therapist's force only in the small-error area. When the error is large and the velocity is small, the therapist's force can be represented by a negative damping system, which is reflected in Fig. 1(b). Fig. 1(b) also shows that the damping will increase to suppress large velocities. These properties are determined by the therapist's demonstrations. Three different types of the experiments are conducted. In these experiments, values of $\bar{A}$,



(a): human stiffness function $k_h(e, \dot{e})$



(b): human damping function $d_h(e, \dot{e})$

Fig. 1. Results for human-impedance learning. (a) Learning results for human-stiffness function $k_h(e, \dot{e})$. (b) Learning results for human-damping function $d_h(e, \dot{e})$.

Q and α are chosen as

$$\begin{cases} \bar{A} = \begin{bmatrix} 0 & 1 \\ 2 & 10 \end{bmatrix} \\ Q = I_{2 \times 2} \\ \alpha = 0.5 \end{cases} \quad (31)$$

which are determined by trial-and-errors in simulations.

In the first type, we evaluate the proposed method when there is no patient force. This type directly relates with the system (6). The experiment results are shown in Fig. 2. As we can see in Fig. 2(c) and Fig. 2(d), the whole process can be divided into three stages, i.e., the initial stage, middle stage and final stage. In the initial stage, the therapist behaves like a negative damping system to rapidly increase the velocity. In the middle stage, the spring-like force appears and the damping will increase to prevent the large velocities. In the final stage, the error is small, and the large positive stiffness and damping will appear to stabilize the system. Thus, the positive spring-damping system only appears in the final stage. These trajectories are consistent with the human-impedance functions shown in Fig. 1.

Another interesting phenomenon is that the complementary input u almost keeps 0 through entire motions, which implies

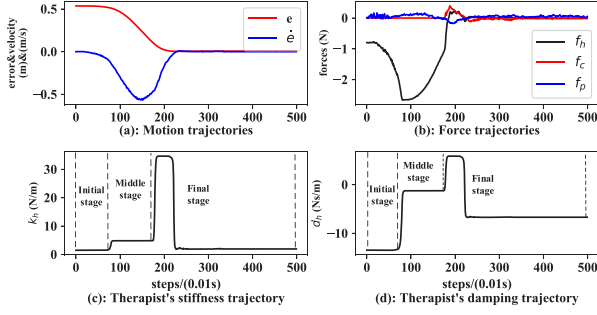


Fig. 2. Robot-rehabilitation experiment results for type1. (a) Error and velocity trajectories. (b) Force trajectories. (c) Therapist's stiffness trajectory. (d) Therapist's damping trajectory.

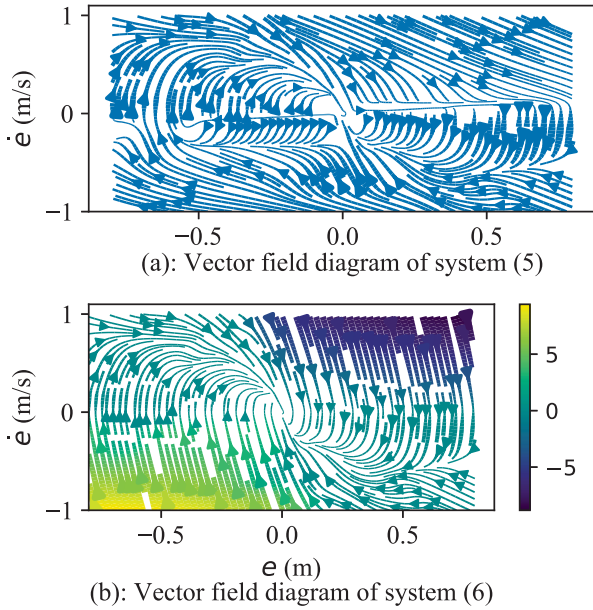


Fig. 3. Vector field diagrams. (a) Vector field diagram of the system (5). (b) Vector field diagram of the system (6).

that the system (6) nearly inherits the optimal property from the demonstration learning. To show this phenomenon more clearly, we plot the vector field diagrams of the systems (5) and (6) as Fig. 3. It can be found from Fig. 3(a) that the demonstration-learning algorithm performed poorly in the area far away from the demonstration data set, such as areas with negative $(e, \dot{e})$ and positive $(e, \dot{e})$. As a result, the systems (5) will be unstable in these areas. This issue can be solved by introducing the complementary input u into the system, which is reflected in Fig. 3(b). The color in the Fig. 3(b) represents the value of the complementary input u . It can be found that the unstable area is corrected by u while the original stable area is not affected, which is the expected situation considering both the optimality and stability.

In the second type of experiments, the patient is also asked to perform the goal-directed motion. Experiment results are shown in Fig. 4. As we can see, the system is still stable without serious corrections. In the third type, an un-normal case caused

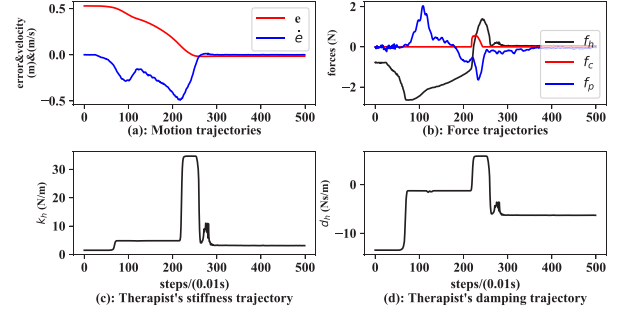


Fig. 4. Robot-rehabilitation experiment results for type2. (a) Error and velocity trajectories. (b) Force trajectories. (c) Therapist's stiffness trajectory. (d) Therapist's damping trajectory.

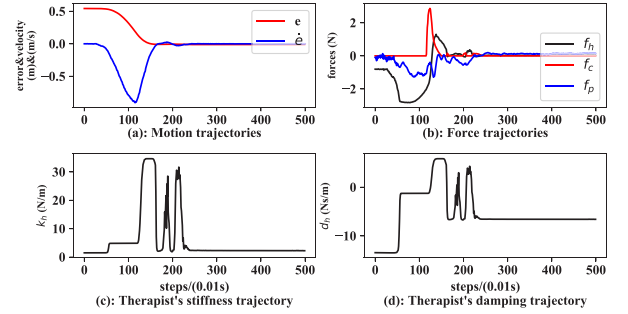


Fig. 5. Robot-rehabilitation experiment results for type3. (a) Error and velocity trajectories. (b) Force trajectories. (c) Therapist's stiffness trajectory. (d) Therapist's damping trajectory.

by the patient's unsuitable force is considered. The experiment result is shown in Fig. 5. The unsuitable patient force causes the large velocity which is judged as the unstable behavior. Thus, an opposite complementary input u is applied to decrease the velocity and stabilize the system.

B. Three-Dimensional Trajectories Reproduction

In this subsection, three-dimensional trajectory reproduction experiments are conducted. Similar to the robotic-assisted rehabilitation experiment, a human is firstly asked to drag the robot to complete some three-dimensional trajectories, and then the robot is required to reproduce these trajectories by using the proposed approach. We consider two different cases, where the robot products trajectories from areas near and away from the demonstration data set, respectively. Moreover, in the first case, the proposed approach is compared with a neural network (NN)-based demonstration learning approach in which the human impedance ground-truth information need to be collected. In these experiments, we decouple the three dimensions and deal with each dimension separately, that is we assume that matrices M , D , D_h and K_h in (5) are diagonal. For each dimension, we use the same parameters as those in the robotic-assisted rehabilitation experiments.

Experiment results for the first case are shown in Fig. 6, where solid lines and dot lines in Fig. 6(a), (b) represent the reproduction trajectories and demonstration trajectories, respectively. Labels, i.e., the human impedance ground-truth

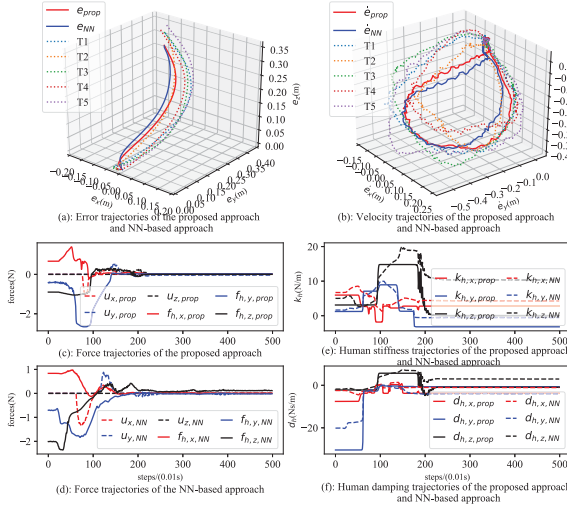


Fig. 6. Comparative results of the proposed approach and the NN-based approach. (a) Error trajectories of the proposed approach and the NN-based approach. (b) Velocity trajectories of the proposed approach and the NN-based approach. (c) Force trajectories of the proposed approach. (d) Force trajectories of the NN-based approach. (e) Human stiffness trajectories of the proposed approach and the NN-based approach. (f) Human damping trajectories of the proposed approach and the NN-based approach.

information used in the NN-based approach, are collected by using the estimation method proposed in [24]. As we can see, the start area of the reproduction trajectories is near with the demonstration area, and reproduction trajectory shapes of the proposed and NN-based approaches are both similar with the demonstration trajectories, which validates the effectiveness of these two demonstration learning approaches. Moreover, we can find that compared with the NN-based approach, the trajectory shape of the proposed approach is more close to demonstration shapes. The reproduction-inaccuracy of the NN-based approach might be caused by the unavoidable estimation error during the label collecting process. Fig. 6(c) and Fig. 6(d) show the force trajectories of the proposed approach and NN-based approach, respectively. We can find that the corresponding force trajectories of these two approaches are similar since these approaches derive similar variable impedance laws (reflected in Fig. 6(e) and Fig. 6(f)). However, the complementary input u_x , u_y and u_z which are used to correct the unstable reproduction behaviors is larger in NN-based approach, which implies that the proposed approach is more stable than NN-based approach. This advantage may benefit from the special form of the GRM result.

Experiment results for the second case are shown in Fig. 7(a) and Fig. 7(b). We can find that if we choose start area away from the demonstration trajectories, the shape of the reproduction trajectories will be different from these of the demonstration trajectories. This phenomenon is common for learning-based algorithms and usually served as a performance index for these algorithms. Since the experiments are conducted in three-dimensional space and the inputs for human-impedance functions are error-velocity pairs, this phenomenon can be serious. For example, as in 7(a), although the error may be close to the demonstration trajectories in a certain period of time,

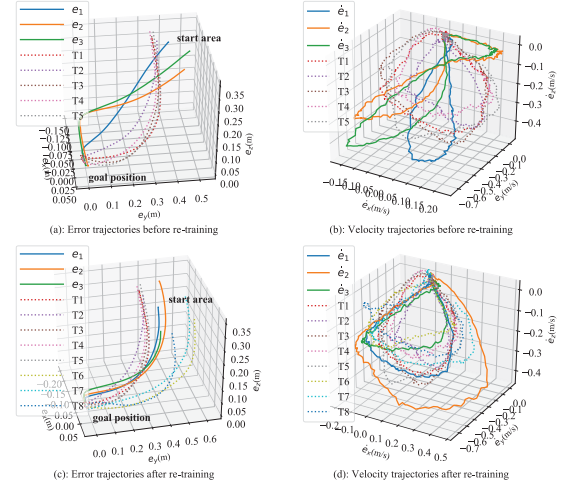


Fig. 7. Three-dimensional trajectory reproduction in area away from the demonstration data set. (a) Error trajectories before re-training. (b) Velocity trajectories before re-training. (c) Error trajectories after re-training. (d) Velocity trajectories after re-training.

the velocity trajectories are far away from the demonstration trajectories in this time. Thus, the estimated human-impedance functions (human force) will perform poorly, which further causes more serious scenes. The issue can be solved by adding additional demonstration trajectories. As we can see in 7(c) and Fig. 7(d), three additional demonstration trajectories, i.e., T6, T7 and T8, are added into the demonstration data set, and we retrain the human-impedance functions on the new demonstration data set. The newly reproduced trajectories are consistent with the demonstration trajectories.

V. CONCLUSION

In this paper, we presented a novel approach to design an optimal and stable VIC algorithm, where the optimality and stability were achieved by using the GMM/GMR-based demonstration learning algorithm and Lyapunov function-based method, respectively. In the demonstration learning, the special formulation of the GMR result was utilized to fit the human-impedance functions without using the ground-truth information. To ensure the stability, we introduced a complementary input into the learned second-order impedance system and designed it by solving a convex optimization problem with Lyapunov function-related constraints. We gave the criterion and method to design the Lyapunov function to ensure the feasibility of the optimization problem, and derived an analytical solution. Finally, the effectiveness of the proposed approach was validated by two practical experiments.

In this paper, a typical quadratic Lyapunov candidate function that satisfies the feasibility criterion given in Proposition 1 was presented, and function parameters were determined by trial-and-errors in simulations. However, both the function formulation and function parameters are not optimal for the demonstration data set, which may decrease the reproduction accuracy of the proposed demonstration learning approach. Thus, determining an optimal Lyapunov candidate function satisfying the

feasibility criterion by using learning or optimization approaches will be considered in our future works.

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